

Dr. Pranav Chakra Varthy Bommarasipeta

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Summary

Civil Engineer with 4 years of industry experience in bridge construction and large-scale infrastructure projects, complemented by a PhD in Civil Engineering at the University at Buffalo. Skilled in structural analysis, finite element modeling, and seismic design using ACI and AISC codes. Proficient in Python and MATLAB for numerical simulations and signal processing. Maintains an active research portfolio with live browser-based structural simulations at <https://innerreality.github.io/RockingDynamicsSite>

Professional Experience

Graduate Research Assistant | Teaching Assistant

Aug 2023 — Aug 2026

University at Buffalo

Buffalo, NY

- Investigated the effects of protective device for electrical transformer bushings in mitigating rupture during earthquake .
- Developed FEM model of transformer in ABAQUS and wrote python scripts for automation and debugging.
- Developed analytical model for hysteretic spring action of protective device and bushing's rocking behavior in MATLAB.
- Conducted shake table tests of life-sized bushing and processed sensor data to fine-tune the numerical models.
- Developed AutoCAD and Autodesk Inventor models for fabrication to repurpose motion table for mini shake table tests.
- Taught civil engineering courses - Statics (Fall 2023, Fall 2025) and Structural Analysis (Summer 2026). Led sessions focused on questioning to build students' ability to examine and challenge their own fundamental assumptions.

Sr. Engineer - Construction Methods

Jul 2019 — Jun 2023

L&T Construction

Chennai, India

- Planned bridge construction and delivered staging plans, fabrication drawings and design documents.
- Coordinated with on-site team for deliverables and immediate response to high-priority technical challenges.
- Contributed to large project packages in India such as Eastern Dedicated Freight Corridor, Mumbai Ahmedabad High-Speed Rail and Chennai Metro Rail (Phase-II) worth \$250 Million (as of 2020)
- Presented construction plans to government authorities to get approval for bridge construction over existing railway line.
- Optimized reuse of temporary structures across sites in EDFC project and saved 2 months of time and about 1 million USD compared to the initial plan.

Graduate Research Assistant

Jul 2018 — Jul 2019

Indian Institute of Technology, Gandhinagar

Gujarat, India

- Studied the dynamic properties of confined masonry buildings and proposed force-based design recommendations for these structures against seismic action.
- Conducted ambient vibration tests on nine (G+3) hostel buildings at IIT Gandhinagar built using confined masonry; performed frequency domain analysis in MATLAB to obtain frequencies and mode shapes.
- Proposed empirical formulas for base shear, frequency, and damping estimates as codal provisions, addressing a gap in Indian and international seismic standards.

Publications

- Abdelrahman Elmeliegy, Jon Bender, Pranav Bommarasipeta, Vidhi Solanki, Mettupalayam Sivaselvan, Aisha Mckee, Bjorn Vaagensmith, and Chandrakanth Bolisetti. Novel Decoupler Solution for Transformer Bushing: Simulation and Experimental Confirmation (In Draft)
- Chakra-Varthy, Pranav & Basu, Dhiman. (2021). Natural period and vertical distribution of base shear in confined masonry buildings using ambient vibration test. Bulletin of Earthquake Engineering. 10.1007/s10518-021-01046-8.
- Kafie, Jackeline & Keating, Peter & Chakra-Varthy, Pranav & Correia, José & De Jesus, Abilio. (2018). Stress distributions and crack growth in riveted lap joints fastening thick steel plates. Engineering Failure Analysis. 91. 10.1016/j.engfailanal.2018.04.048.

Conferences

- Sai Niranjana Ramachandran, Pranav Chakravarthy Bommarasipeta. “Consensus Under Noise: A Unified Analysis of Social Learning in Discrete and Continuous Belief Spaces” Late-Breaking Research, Proceedings of the 8th Annual Learning for Dynamics and Control Conference (L4DC), 2026.

Projects

Rocking Dynamics (innerreality.github.io/RockingDynamicsSite)

- Developed an interactive web-based resource presenting the derivation of the rocking ODE using a smooth sign approximation to model rigid rocking behavior, spanning three chapters with accompanying browser simulations.
- Implemented chapter-level mini simulations and a full-scale ODE solver simulation for each chapter, porting numerical solvers from MATLAB to JavaScript for live in-browser execution.
- Built with JavaScript to make structural dynamics concepts accessible through interactive visualization

Education

PhD in Civil Engineering

University at Buffalo

Buffalo, NY

Aug 2023 — Aug 2026

- Recipient of Graduate Fellowship from University at Buffalo from Fall-2023 to Spring-2026

Master of Technology in Civil Engineering (Integrated B.Tech-M.Tech Dual Degree)

Indian Institute of Technology, Gandhinagar

Gujarat, India
Aug 2018 — Jul 2019

Bachelor of Technology in Civil Engineering (Minors in Computer Science)

Indian Institute of Technology, Gandhinagar

Gujarat, India

Aug 2014 — Jul 2018

- Received Academic Excellence Scholarship for the year at IITGN (2016–2017)
- Awarded Dean’s list for Academic Excellence at IITGN: 2015–2016 (Semester-I), 2014–2015 (Semester-I)
- Recipient of S.P Mehrotra Scholarship for the years at IITGN (2015–2018)

Technical Skills

- **Design Tools:** ABAQUS, LARSA 4D, SAP 2000, Staad Pro, MathCAD, AutoCAD, Autodesk Inventor
- **Programming Languages:** Python, MATLAB
- **Software & Tools:** Microsoft Office, Linux commands, Git & Agentic workflows